

Soham Belurgikar

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EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science (MCS)

Urbana-Champaign, IL
Aug. 2026 - May 2028 (Expected)

Sardar Patel Institute of Technology

B.Tech in Computer Engineering; CGPA: 9.21/10.0

Mumbai, India
Aug. 2019 - May 2023

EXPERIENCE

Barclays

Software Engineer (BA4)

Pune, India
Apr. 2025 - Jul. 2026

- Built a distributed comparison framework for 10+ node Spark clusters, isolating race conditions and cutting regression verification time across runtime versions by 50%.
- Architected an on-demand synthetic data platform using React, Spring Boot, and Python, generating 1M+ records to enable load testing without production PII.
- Automated data reconciliation and validation pipelines across staging and production, reducing manual verification effort by approximately 40%.

Software Engineer (BA3)

Jul. 2023 - Mar. 2025

- Modernized core distributed data pipelines by migrating batch workloads to Spark 3.x on open-source Hadoop (HDFS), eliminating runtime bottlenecks with zero production downtime.
- Engineered automated CI/CD pipelines via Jenkins and Groovy, cutting deployment cycle times by 35% across 15+ distributed artifacts and staging environments.
- Benchmarked and tuned Spark SQL query plans across 10+ TB financial datasets, resolving execution bottlenecks while preserving 100% data integrity.

Summer Intern (BA3)

Jun. 2022 - Jul. 2022

- Developed a Python tool that automated cross-system dataset reconciliation and generated structured discrepancy reports for millions of records, packaged in Docker and integrated into a Jenkins pipeline, enabling the team to spot data mismatches quickly and cut manual review effort.

PROJECTS

FrameSeek - Multimodal Video Search Engine

Python, PyTorch, scikit-learn, Go, PostgreSQL, LangChain, LangGraph, Docker, AWS

- Built an end-to-end multimodal search platform combining transcripts, OCR, and visual embeddings across 100+ hours of video; served results via a Go REST API at 500 ms p95 search latency.
- Trained a logistic regression relevance scorer in scikit-learn, boosting query Success@5 by +8 percentage points over OCR/transcript baselines on 300 held-out queries (temporal IoU ≥ 0.5).
- Engineered a fault-tolerant ingestion pipeline on AWS S3 and Batch with idempotent retries and stage checkpoints, achieving zero data loss across 100+ GB of raw video feeds.
- Implemented conversational multi-turn retrieval using LangGraph and custom LangChain agents to resolve follow-up context and return timestamp-linked playback results.

StreamForge - Distributed Real-Time Feature Store & Pipeline

Go, Python, Apache Kafka, Redis, gRPC, Docker, PostgreSQL

- Designed a distributed streaming pipeline in Go and Kafka processing 25,000+ events/sec to compute and serve real-time feature vectors for downstream ML inference models.
- Implemented a Raft-based consensus mechanism in Go for fault-tolerant state replication across a 5-node cluster, ensuring consistent state updates during network partitions.
- Exposed low-latency feature retrieval via gRPC microservices with Redis caching, sustaining <12 ms p99 read latency under simulated high concurrency.
- Containerized the multi-service architecture with Docker Compose and configured Prometheus/Grafana alerting, maintaining 99.9% uptime across cluster test runs.

TECHNICAL SKILLS

Languages: Python, Go, Java, TypeScript, C++, SQL, C, Groovy, Bash

Distributed systems & data: Apache Spark, Apache Kafka, Hadoop (HDFS), PostgreSQL, Redis

ML & retrieval: PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph, pandas, NumPy

Cloud & DevOps: AWS (S3, Batch, EC2), Docker, Jenkins, CI/CD, AI-assisted development, Linux, Git, GitLab

Frameworks & protocols: Spring Boot, Django, React, Angular, Node.js, REST APIs, gRPC

PUBLICATIONS

[Google Scholar](#)

- **A Novel Approach to Fall Detection System using Computer Vision.** Co-author, ICACCS 2023.
- **Street Smart: An Intelligent Traffic Management System.** Co-author, INDIACom 2023.